

Balancing trade-offs between biotic and abiotic stress responses through leaf age-dependent variation in stress hormone cross-talk

Matthias L. Berens^a, Katarzyna W. Wolinska^a, Stijn Spaepen^a, Jörg Ziegler^b, Tatsuya Nobori^a, Aswin Nair^{a,1}, Verena Krüler^{a,2}, Thomas M. Winkelmüller^a, Yiming Wang^a, Akira Mine^{a,3}, Dieter Becker^a, Ruben Garrido-Oter^{a,c}, Paul Schulze-Lefert^{a,c,4}, and Kenichi Tsuda^{a,4}

^aDepartment of Plant-Microbe Interactions, Max Planck Institute for Plant Breeding Research, 50829 Cologne, Germany; ^bDepartment of Molecular Signal Processing, Leibniz Institute of Plant Biochemistry, 06120 Halle, Germany; and ^cCluster of Excellence on Plant Sciences, Max Planck Institute for Plant Breeding Research, 50829 Cologne, Germany

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In nature, plants must respond to multiple stresses simultaneously, which likely demands cross-talk between stress-response pathways to minimize fitness costs. Here we provide genetic evidence that biotic and abiotic stress responses are differentially prioritized in *Arabidopsis thaliana* leaves of different ages to maintain growth and reproduction under combined biotic and abiotic stresses. Abiotic stresses, such as high salinity and drought, blunted immune responses in older rosette leaves through the phytohormone abscisic acid signaling, whereas this antagonistic effect was blocked in younger rosette leaves by *PBS3*, a signaling component of the defense phytohormone salicylic acid. Plants lacking *PBS3* exhibited enhanced abiotic stress tolerance at the cost of decreased fitness under combined biotic and abiotic stresses. Together with this role, *PBS3* is also indispensable for the establishment of salt stress- and leaf age-dependent phyllosphere bacterial communities. Collectively, our work reveals a mechanism that balances trade-offs upon conflicting stresses at the organism level and identifies a genetic intersection among plant immunity, leaf microbiota, and abiotic stress tolerance.

combined stress | phytohormone | plant fitness | microbiota | stress trade-off

In nature, plants encounter and must appropriately respond to diverse stresses to survive and reproduce. Different stress signaling pathways are connected, providing regulatory potential to maximize fitness (1). For instance, plants exposed to abiotic stresses such as high salinity and drought often display reduced immune activity (2). As stress responses are costly, prioritization in stress responses would allow plants to allocate more resources to abiotic stress responses and increase plant fitness in the absence of biotic stress (3). However, this prioritization does not explain how plants maintain fitness when simultaneously exposed to biotic and abiotic stresses.

Phytohormones play critical roles in stress responses and prioritization (4). In an evolutionarily conserved mechanism, phytohormone signaling mediated by abscisic acid (ABA) promotes abiotic stress tolerance and suppresses signaling of the biotic stress-related phytohormone salicylic acid (SA). Consequently, plant immunity is lowered during abiotic stresses such as drought and salt stress (5). In *Arabidopsis thaliana*, ABA suppresses expression of the SA biosynthesis gene *SA INDUCTION DEFICIENT 2 (SID2)* (5). ABA application causes proteasome-mediated degradation of the SA receptor *NONEXPRESSOR OF PR GENE 1 (NPR1)* (6). Molecular cross-talk between ABA and SA provides a plausible mechanism for prioritization of abiotic over biotic stress responses. Such a mechanism would provide plants with adaptive potential when abiotic stresses are major threats. However, it is unknown whether this hormonal cross-talk is still advantageous when plants simultaneously encounter biotic and abiotic stresses.

In natural soils, healthy plants are colonized by taxonomically structured microbial communities, collectively called the plant

microbiota (7), which contribute to plant performance under adverse environmental conditions (8–11). SA modulates colonization of the *A. thaliana* root microbiota by specific bacterial taxa, although this was not accompanied by a detectable impact on host survival of the tested SA biosynthesis and signaling mutants (12, 13). In field experiments, drought induced a relative enrichment of multiple lineages of Actinobacteria and Chloroflexi in the rice root microbiota, whereas, in sorghum roots, relative enrichments of Actinobacteria and Firmicutes were observed (14–17). Whether and how cross-talk between biotic and abiotic stress response pathways impacts the assembly of the plant microbiota remains unknown.

Significance

Plants are exposed to conflicting stresses simultaneously in nature. As stress responses are costly, plants likely coordinate these responses to minimize fitness costs. The nature and extent to which plants employ inducible mechanisms to cope with combined physical and biological stresses remains unknown. We identify a genetic mechanism by which leaves of distinct ages differentially control stress-response cross-talk. At the organism level, this mechanism balances stress-response trade-offs to maintain plant growth and reproduction during combined stresses. We also show that this leaf age-dependent stress-response prioritization influences the establishment of plant-associated leaf bacterial communities. This study illustrates the importance of active balancing of stress-response trade-offs for plant fitness maintenance and for interaction with the plant microbiota.

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Data deposition: The data reported in this paper have been deposited in the Gene Expression Omnibus (GEO) database, <https://www.ncbi.nlm.nih.gov/geo> (accession no. [GSE114645](#)) and in the European Nucleotide Archive, <https://www.ebi.ac.uk/ena> (accession no. [PRJEB26793](#)).

¹Present address: Albrecht-von-Haller-Institut für Pflanzenwissenschaften, Georg-August-Universität Göttingen, D-37073 Göttingen, Germany.

²Present address: Institut für Biologie und Biotechnologie der Pflanzen, Westfälische Wilhelms Universität, 48149 Münster, Germany.

³Present address: Ritsumeikan Global Innovation Research Organization, Ritsumeikan University, 525-8577 Shiga, Japan.

⁴To whom correspondence may be addressed. Email: schlef@mpipz.mpg.de or tsuda@mpipz.mpg.de.

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